

Product Line Profitability and Margin Performance Analysis for Nassau Candy Distributor Using Data Analytics and Interactive Visualization

Rikkala Akshitha

Department of Computer Science and Engineering, Malla Reddy Vishwa Vidyapeeth, Hyderabad, India

Abstract—This paper presents a data-driven framework for analyzing product-level profitability and margin performance for Nassau Candy Distributor. It addresses the limitation of sales-only evaluation by combining sales, cost, gross profit, margin, unit economics, contribution, and temporal measures. A transactional dataset containing 10,194 records and 15 distinct products was analyzed using Python, Pandas, NumPy, Matplotlib, Seaborn, and Streamlit. The workflow includes data validation, gross-margin and profit-per-unit calculation, product and division analysis, Pareto profit concentration, cost-versus-sales diagnostics, and monthly margin analysis. The dataset contained no missing values, duplicate records, invalid sales, negative costs, or non-positive unit quantities. Aggregate results were approximately \$141,784 in sales, \$48,341 in cost, \$93,443 in gross profit, and 38,654 units sold, corresponding to an overall gross margin of approximately 65.91%. An interactive Streamlit dashboard was developed with date, division, product, and margin filters. The framework demonstrates how transactional data can be transformed into practical profitability intelligence for pricing, cost management, and product portfolio review.

Index Terms—Product Profitability; Gross Margin; Data Analytics; Pareto Analysis; Cost Diagnostics; Streamlit; Business Intelligence

1. INTRODUCTION

Distribution businesses manage product portfolios in which revenue, unit volume, cost, and margin can differ substantially across products. A product that appears successful when ranked by sales may contribute relatively little profit after its associated cost is considered. Conversely, a lower-sales product can have stronger economics when its margin and profit per unit are favorable.

This study develops a reproducible analytical framework for Nassau Candy Distributor that transforms transactional sales data into product and division profitability intelligence. The work combines descriptive analytics, financial metric calculation, product segmentation, Pareto analysis, cost diagnostics, temporal analysis, and interactive visualization.

The research question is: How can product-level sales, cost, and gross-profit data be transformed into an interpretable framework that identifies profitability drivers, margin risks, and profit concentration within a distributor's product portfolio?

2. LITERATURE REVIEW

Product profitability analysis is important because sales and margin distributions do not necessarily have the same structure. Research on retail gross-margin distributions shows that gross-margin outcomes can differ from sales outcomes, supporting the use of margin-based analysis alongside revenue measures [1].

Pareto analysis has been applied to product and distribution problems to identify a relatively small group of products responsible for a large share of profit or other operational outcomes [2]. In a profitability setting, cumulative contribution analysis can reveal dependency on a limited number of products.

Business data analytics provides a process for transforming raw data through preparation, analysis, visualization, and interpretation into information that can support decisions [3]. Interactive dashboards extend this approach by allowing users to investigate subsets of the data rather than relying only on static reports.

The present study combines these ideas into a single product-profitability workflow using Python analytics and a Streamlit decision-support interface.

3. PROBLEM STATEMENT AND OBJECTIVES

The organization requires greater visibility into which product lines generate the highest gross margin, whether high-sales products are financially attractive, how profitability varies across divisions, and which products may represent margin risk. Without such visibility, pricing, sourcing, promotion, and portfolio decisions may rely mainly on aggregate sales or intuition.

The objectives are to validate the transactional dataset; calculate product profitability metrics; rank products by gross profit and gross margin; compare division-level performance; classify high-sales/low-margin and low-sales/low-profit products; quantify profit concentration through Pareto analysis; diagnose cost-heavy and margin-poor products; examine monthly margin behavior; and develop an interactive dashboard.

4. DATASET AND PREPROCESSING

The dataset contains 10,194 transactional records and 18 original fields covering order information, customer geography, product attributes, sales, units, gross profit, and cost. The product portfolio contains 15 unique products across Chocolate, Sugar, and Other divisions. Order dates span approximately January 2024 through December 2025.

Data validation included missing-value checks, duplicate checks, financial consistency checks, unit validation, and date conversion. The dataset contained zero missing values and zero duplicate rows. There were no sales values less than or equal to zero, no negative cost values, and no non-positive unit quantities.

Gross profit was independently validated using $\text{Gross Profit} = \text{Sales} - \text{Cost}$. The maximum absolute difference between the calculated and supplied values was approximately 7.1×10^{-15} , consistent with floating-point representation.

Shipping Days was calculated as Ship Date minus Order Date. The supplied dates produced unusually long intervals; therefore, the dates were not artificially altered without source verification.

5. METHODOLOGY

The analytical workflow consists of data preparation, metric calculation, aggregation, segmentation, diagnostic analysis, and dashboard development. Pandas supported data manipulation and grouping, NumPy supported calculations and classification logic, Matplotlib and Seaborn supported visualization, and Streamlit provided the interactive application layer.

$\text{Gross Margin (\%)} = (\text{Gross Profit} / \text{Sales}) \times 100$. $\text{Profit per Unit} = \text{Gross Profit} / \text{Units}$. $\text{Revenue Contribution} = \text{Product Sales} / \text{Total Sales}$. $\text{Profit Contribution} = \text{Product Gross Profit} / \text{Total Gross Profit}$.

For temporal analysis, transactions were aggregated by calendar month. Monthly gross margin was calculated from monthly gross profit and monthly sales. Margin volatility was represented by the standard deviation of monthly gross margin.

Product performance was segmented using median sales and median gross margin into High Sales–High Margin, High Sales–Low Margin, Low Sales–High Margin, and Low Sales–Low Margin. These are analytical flags rather than automatic business decisions.

Pareto analysis ranks products by gross profit, calculates each product's contribution to total gross profit, and then calculates cumulative contribution to identify profit concentration.

Cost diagnostics compare cost with sales and margin. Products with relatively high cost and weak margin, or high sales combined with weak margin, are flagged for further investigation. Potential actions include pricing review, cost renegotiation, promotion review, or portfolio rationalization.

6. RESULTS

At the portfolio level, total sales were approximately \$141,784, total cost was approximately \$48,341, gross profit was approximately \$93,443, and total units sold were 38,654. The resulting aggregate gross margin was approximately 65.91%.

Product-level analysis ranked products using total gross profit and gross margin, while profit per unit provided an additional unit-economics perspective. The resulting classifications distinguish products that generate revenue from products that generate sustainable profitability.

Division-level aggregation used total sales, cost, gross profit, units, and aggregate gross margin. Aggregate margin was calculated as total division gross profit divided by total division sales rather than an unweighted mean of transaction margins.

Pareto analysis provides a portfolio concentration view through cumulative gross-profit contribution. A concentrated curve indicates that a relatively small number of products account for a large share of profit and may create dependency risk.

Cost-versus-sales diagnostics complement profitability rankings because a product can generate high sales while retaining a relatively small portion as gross profit when cost is also high.

7. STREAMLIT DECISION-SUPPORT DASHBOARD

An interactive Streamlit application was developed with product profitability overview, product-level analysis, division performance, cost and margin diagnostics, profit concentration analysis, and risk indicators.

The interface provides a date-range selector, division filter, product selector, minimum gross-margin slider, product search, and filtered CSV download. This allows users to investigate specific subsets without changing the underlying analytical code.

The dashboard is intended as a decision-support layer rather than a replacement for financial systems. It provides a consistent interface for exploratory profitability analysis.

8. DISCUSSION

The results support the central premise that sales volume alone is insufficient for product portfolio evaluation. The overall portfolio margin of approximately 65.91% is useful as a high-level KPI, but it does not reveal which individual products create or dilute profitability.

The combination of gross profit, gross margin, and profit per unit provides complementary views. Gross profit identifies absolute profit contribution, gross margin describes efficiency relative to sales, and profit per unit provides a unit-level economic measure.

Division analysis adds organizational context, while Pareto analysis adds concentration context. Cost diagnostics then provide a mechanism for identifying products whose financial behavior deserves deeper review.

The dashboard improves accessibility by allowing non-programming users to change filters and inspect the corresponding analytical subset.

9. LIMITATIONS

The study is based on the supplied transactional dataset and reflects only the variables available in it. Gross profit should not be interpreted as complete net profitability because additional operating expenses, taxes, overheads, and other financial items may not be represented.

The analytical flags identify products for investigation but do not prove causality. Any pricing, sourcing, or discontinuation decision requires business context not contained in the dataset.

The supplied shipment-date structure produced unusually large order-to-ship intervals. The dates were not artificially corrected; a verified source system should be consulted before using those intervals for logistics claims.

10. CONCLUSION AND FUTURE WORK

This study developed an end-to-end framework for Product Line Profitability and Margin Performance Analysis for Nassau Candy Distributor. The framework validates transactional data, calculates profitability KPIs, compares products and divisions, evaluates profit concentration, diagnoses cost-related margin risk, analyzes temporal behavior, and exposes results through an interactive Streamlit dashboard.

The portfolio generated approximately \$141,784 in sales and \$93,443 in gross profit from 38,654 units, producing an aggregate gross margin of approximately 65.91%. These aggregate results provide a useful baseline, while product-level and division-level analysis adds the detail needed to investigate profitability differences.

Future work can extend the framework with customer-level profitability, product-demand forecasting, margin prediction, statistical hypothesis testing, pricing scenario optimization, supplier-cost analysis, and automated dashboard refresh.

Table 1. Portfolio-Level KPI Summary

KPI	Value
Transactions	10,194
Unique Products	15
Total Sales	~\$141,784
Total Cost	~\$48,341
Gross Profit	~\$93,443
Units Sold	38,654
Aggregate Gross Margin	~65.91%

REFERENCES

[1] J. S. Strong, "The distribution of retail gross margin: analysis and implications," International Review of Retail, Distribution & Consumer Research, vol. 34, no. 5, 2024.

[2] "A multi-method approach for reducing operational wastes in distribution warehouses," International Journal of Production Economics, 2022, doi: 10.1016/j.ijpe.2022.108705.

[3] IEEE Technology Navigator, "Data Analytics," IEEE.

[4] IEEE Author Center Conferences, "Structure Your Paper," IEEE.

[5] IEEE Author Center Conferences, "Types of IEEE Conference Papers," IEEE.

Publication note—This is an academic research-paper draft based on the completed project analysis. It is not evidence of acceptance or publication by IEEE or another venue. Check the target venue template and requirements before formal submission.